

Strong Gibbs convergence for every ℓ^2 initial condition

Candidate full answer to Remark (2) in arXiv:2011.00457

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Status: candidate full answer. Under the hypotheses of the source's Main Theorem, the untruncated master-equation semigroup converges strongly to its Gibbs-mode projection for every initial condition in ℓ^2 . A quantitative head-tail estimate is available, but operator-norm convergence is impossible at every positive time.

1 Source question and answer

Bögli and Vuillermot [1] study a compact, nonnormal generator A on $\ell_{\mathbb{C}}^2$. Under the gap-control hypotheses of their Main Theorem, its eigenvectors $(\widehat{p}_k)_{k \geq 1}$ form a Schauder basis and

$$e^{tA}x = \sum_{k=1}^{\infty} \langle x, \widehat{q}_k \rangle e^{t\nu_k} \widehat{p}_k \quad (t \geq 0) \quad (1)$$

in norm. Here $\nu_1 = 0$, while

$$\nu_2 < \nu_3 < \cdots < 0, \quad \nu_k \longrightarrow 0, \quad (2)$$

and $\widehat{p}_1$ is a scalar multiple of the Gibbs vector.

Remark (2) after their corollary proves the limit below for $x \in \ell^1$ and asks whether it holds for every $x \in \ell^2$. Define the bounded coordinate projection

$$\Pi x = \langle x, \widehat{q}_1 \rangle \widehat{p}_1.$$

Theorem 1 (Full ℓ^2 convergence). *Under the hypotheses of the source's Main Theorem,*

$$\lim_{t \rightarrow \infty} \|e^{tA}x - \Pi x\|_2 = 0 \quad \text{for every } x \in \ell_{\mathbb{C}}^2. \quad (3)$$

More precisely, let

$$P_N x = \sum_{k=1}^N \langle x, \widehat{q}_k \rangle \widehat{p}_k, \quad K = \sup_N \|P_N\| < \infty.$$

For every $N \geq 2$ and $t \geq 0$,

$$\|(e^{tA} - \Pi)x\|_2 \leq 2K e^{t\nu_N} \|x\|_2 + (1 + K) \|(I - P_N)x\|_2. \quad (4)$$

On the other hand,

$$\|e^{tA} - \Pi\| \geq 1 \quad (t \geq 0), \quad (5)$$

so the convergence is never in operator norm and admits no uniform decay rate on the unit ball.

2 Proof intuition

Coefficientwise convergence in (1) is not enough by itself: the eigenvectors are a Schauder basis, not asserted to be unconditional. The missing control comes from the order of the eigenvalues. For fixed t , the non-Gibbs multipliers

$$e^{t\nu_2} \leq e^{t\nu_3} \leq \dots \longrightarrow 1$$

are monotone, so their total variation is uniformly bounded. Abel summation transfers that scalar variation bound to the basis multiplier through the uniformly bounded partial-sum projections P_N . The resulting operators are uniformly bounded in time. They vanish on each finite basis span as $t \rightarrow \infty$, and density finishes the proof.

The same calculation separates a finite head, which decays at rate $e^{t\nu_N}$, from an arbitrarily small basis tail. Yet high-index eigenvalues are arbitrarily close to zero, and their multipliers are arbitrarily close to one at each fixed time; this gives the sharp norm obstruction (5).

3 Abel multiplier lemma

Lemma 1. *Let (u_k) be a Schauder basis of a Banach space, with partial-sum projections P_N and $K = \sup_N \|P_N\|$. If (a_k) has finite total variation and $a_k \rightarrow L$, then its basis multiplier satisfies*

$$M_a = LI + \sum_{n=1}^{\infty} (a_n - a_{n+1})P_n, \quad \|M_a\| \leq |L| + K \sum_{n=1}^{\infty} |a_n - a_{n+1}|. \quad (6)$$

Proof. For every finite basis expansion, the coefficient of u_k on the right of (6) is

$$L + \sum_{n=k}^{\infty} (a_n - a_{n+1}) = a_k.$$

The series of operators converges absolutely by the variation hypothesis, and the norm estimate follows from $\|P_n\| \leq K$. Density extends the identity to the whole space. $\square$

4 Proof of Theorem 1

Put $D_t = e^{tA} - \Pi$. Relative to the basis $(\widehat{p}_k)$, its scalar multipliers are

$$a_1(t) = 0, \quad a_k(t) = e^{t\nu_k} \quad (k \geq 2).$$

By (2), $a_k(t)$ increases for $k \geq 2$ and tends to 1. Consequently

$$\sum_{n=1}^{\infty} |a_n(t) - a_{n+1}(t)| = a_2(t) + (1 - a_2(t)) = 1.$$

Lemma 1 gives the time-uniform bound

$$\|D_t\| \leq 1 + K. \quad (7)$$

To obtain the stated estimate, truncate the multiplier after index N . The finite scalar sequence

$$0, a_2(t), \dots, a_N(t), 0, 0, \dots$$

has total variation $2a_N(t) = 2e^{t\nu_N}$ and limit zero. A second application of Lemma 1, followed by (7), yields

$$\begin{aligned}\|D_t x\|_2 &\leq \|D_t P_N x\|_2 + \|D_t(I - P_N)x\|_2 \\ &\leq 2Ke^{t\nu_N}\|x\|_2 + (1 + K)\|(I - P_N)x\|_2,\end{aligned}$$

which is (4).

Given x and $\varepsilon > 0$, first choose N so large that the tail term in (4) is below $\varepsilon/2$. Since $\nu_N < 0$, next choose t so large that the head term is below $\varepsilon/2$. This proves (3).

Finally, $D_t \hat{p}_k = e^{t\nu_k} \hat{p}_k$ for $k \geq 2$. Hence

$$\|D_t\| \geq \sup_{k \geq 2} e^{t\nu_k} = 1$$

because $\nu_k \rightarrow 0$. This proves (5) and completes the proof.

5 Scope and literature check

The theorem answers exactly the ordinary- ℓ^2 question under all the assumptions used by the source to obtain its Schauder-basis spectral theorem. It does not remove those hypotheses. The estimate (4) gives an orbit-dependent convergence mechanism, while (5) explains why no spectral-gap-style uniform exponential estimate can exist.

Searches through 12 August 2026 located the 2022 journal version of the source and a later grand-canonical paper by the same authors in a different, weighted self-adjoint sequence space, but no statement of this exact nonnormal ordinary- ℓ^2 conclusion. Novelty should nevertheless be checked in specialist citation databases before publication.

References

- [1] S. Bögli and P.-A. Vuillermot, *A spectral theorem for the semigroup generated by a class of infinitely many master equations*, Acta Appl. Math. 178 (2022), article 4; arXiv:2011.00457.